

Biochemistry

Sem 3

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===== SOURCE_FILE: sem 3/Biochemistry/PYQs/PYQ Dec 2022.pdf =====

CATEGORY: PYQs

STATUS: ok

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Reg. No.

B.Tech. DEGREE EXAMINATION, DECEMBER 2022

Third Semester

18BTC101J - BIOCHEMISTRY

(For the candidates admitted from the academic year 2020-2021 to 2021-2022)

Note:

(i) Part - A should be answered in OMR sheet within first 40 minutes and OM R sheet should be handed

over to hall invigilator at the end of 40th minute.

(ii) Part - B should be answered in answer booklet.

Time: 2½ Hours Max. Marks: 75

PART - B (5 × 10 = 50 Marks)

Answer ALL Questions

Marks

BL

CO

PO

26. a. Write an essay on the different types of polysaccharide. Add a short note on lectins.

10 3

1 1,1

3

(OR)

b. Explain the structure and organization of proteins, and mention the chemical bonds involved in it.

10 3

1 1,1

3

27. a. Explain the biochemical reactions that occur in TCA cycle and describe how the citric acid cycle is regulated.

10 4

1 1,4

,13

(OR)

b. Classify the diabetes mellitus and list out the complications associated with them. Add a note on insulin release in the pancreas.

10 4

1 1,4

,13

28. a. Write an essay on the important biomolecules that are derived from amino acids.

10 3 3 1,4

,14

(OR)

b. Solve the step by step process of how the two nitrogen from amino acids are removed as urea in our body.

10 3 3 1,4

,14

29. a. List out the steps and identify the repeated steps of fatty acids degradation.

10 4 4 1,4

,15

(OR)

b. Describe the pathway through which purine nucleotide synthesis occurs.

10 4 4 1,4

,15

30. a. Describe the key components of the electron transport chain, and add notes on how they are arranged?

10 4 5 4,5

,15

(OR)

b. Describe how the light reactions and the Calvin cycle are coordinated.

10 4 5 4,5

,15

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===== SOURCE_FILE: sem 3/Biochemistry/PYQs/PYQ May 2023.pdf =====

CATEGORY: PYQs

STATUS: ok

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Note: (i) (ii) Time: 3 Hours

Reg. No. R

Part -A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute. Part -B and Part -C should be answered in answer booklet.

(A) Proteins

B.Tech. DEGREE EXAMINATION, MAY 2023 Second Semester (For the candidates admitted from the academic year 2022-2023)

1. A phosphodiester bond is present between (A) Fatty acids in a diglyceride (C) Amino acids in proteins

(C)

(C) Carbohydrates

22

21BTC101T -BIOCHEMISTRY

PART -A (20 x 1 = 20Marks) Answer ALL Questions

3. a-D-glucose and B-D-glucose are (A) Epimers

2. Which of the following macromolecules have the general molecular formula $(CH_2O)_n$?

Keto-ALDO pairs

(A) Glycogen (C) Fructose

(A) Stromal matrix

(B) Monosaccharides in carbohydrates (D) Nucleotides in DNA

(C) Photo system -I

(B) Lipids

o37

(D) Nucleic acids

4. The phospholipid that prevents the adherence of inner surface of lungs (A)

Phosphoinositides (C) Cephalin

(B) Anomers

5. Which of the following has the highest standard reduction potential? (A) NAD (C) Oxygen

(D) Constitutional isomers (B) Lecithin (D) Plasmogens

6. The following is the most preferred source of energy for the brain (B) FMN (D) Pyruvate

(B) Glucose (D) Fatty acids 7. Light independent reactions in photosynthesis takes place at

(B) Thylakoid lumen (D) Photo system -II 8. The following uncouples electron transport in

the mitochondrion from the production of chemical energy in the form of adenosine

triphosphate (ATP) (A) Thymine (B) Thermogenin (C) Cytochromes (D) Heme

Max. Marks: 75

Marks BL CO PO

1

1

1

1

2

1

2

1

1 1,4

1 1.4

1 1,4

1 1,4

2 1,3

2 1,3

2 1,3

2 1.3

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9. Which of the following are the major sites for glycogen storage (A) Muscle and brain (C)

Liver and kidney 10. Which of the following is the precursor of glycogen? (A) UDP glucose

(C) Glycerol 3-phosphate of gluconeogenesis 11. Which of the following hormones maintain

blood glucose level by activation (A) Epinephrine (C) Glucagon (A) Oxidative decarboxylation

(C) Dehydration

P'age 2 of 3

(A) Alanine

12. The following reactions generate acetyl CoA from pyruvate

(C) Choline degraded to (A) Fumarate (C) Pyruvate

13. One of the following is NOT an amino acid

- (A) Glutamate
 (A) α-keto glutarate
 (B) Muscle and liver (D) Adipose tissue
 14. The amino acids are said to be ketogenic when the carbon skeleton is finally
 (C) Oxalo acetate
 (B) Malate
 (A) Acetyl CoA
 (D) Leucine and lysine
 (C) Malonyl CoA
 15. The following amino acid do not participate in transamination
 (B) Insulin
 (A) Orotidine monophosphate
 (D) Nor-epinephrine
 (C) Xanthosine monophosphate
 16. The common compound shared by both TCA cycle and urea cycle is
 (B) Reduction
 (A) Aspartate and glutamine (C) Asparagine and leucine 20. Ketone bodies are formed from (A)
 Acetyl CoA (C) Acetoacetate
 (D) Dephosphorylation
 (B) Proline (D) Glycine
 17. The following is the source of all carbon atoms in cholesterol
 (B) Acetyl CoA (D) Succinyl CoA (B) Tryptophan (D) Lysine
 18. The parent nucleotide synthesized in the biosynthesis of purines is
 (B) Fumarate (D) Succinyl CoA (B) Propionyl CoA (D) Acetoacetyl CoA
 19. The amino acids required for the synthesis of pyrimidines are (B) Adenyl succinate (D)
 Inosine monophosphate (B) Alanine and cysteine (D) Methionine and tyrosine (B) Acetone (D)
 Propionyl CoA

1

1

2

2

2

3

1

3 L4

2

14

3 L4

| 4 1.2

| 4

4 1,2

4 1,2

1,2 4

29MF221BTC101T

4

S 1,2

1,2

5 1.2

(C) Alanine

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PART-B (5 x 8 = 40 Marks) Answer ALL Questions) a What are enzymes? Draw the activation energy diagram with and without the enzyme and describe the major classes of enzymes. (OR) b What is pH? Define Bronsted-Lowry acid and base and derive the Henderson-Hasselbalch equation. 22.a. Give an account on photorespiration and explain how C4 plants avoid photorespiration. (OR) b. What is shuttle pathway? Explain one of the shuttle pathways in detail. 23.a. What is diabetes mellitus? Describe briefly about the blood glucose level regulation by insulin. (OR) b. What is glycogenolysis? Which organs carry out this process? Briefly describe about glycogenolysis. 24. a. Write the salient features of transamination of amino acids and describe the mechanism of transamination in detail. (OR)

Marks

b. Mention any four amino acid metabolic disorders and describe about phenylketonuria in detail. 25. a. Discuss the steps involved in the β -oxidation of fatty acid. (OR) b. Explain the process by which pyrimidines are synthesized. PART C(1x 15=15 Marks) Answer ANY ONE Questions

8

BL

2.3 1

8

26. Explain the electron transport chain with the help of a diagram and discuss the synthesis of ATP by oxidative phosphorylation with the help of chemiosmotic theory / hypothesis proposed by Peter Mitchell.

2,3

CO

2,3

8

27. Ammonia is a toxic product of nitrogen (amino acid) metabolism and it must be removed from the body. Explain the mechanism of removal of ammonia from body.

2,3

2

2,3

Marks

2

15

2

PO

1 1,4

2

2

1,4

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CO P
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